



# NIU Xuezhhi

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Uppsala, Sweden

## Education

- Uppsala University 2024–2028(Est.)  
Ph.D. Student in Embedded Systems Uppsala, Sweden
- KTH Royal Institute of Technology 2021–2023  
M.Sc. Mechatronics Stockholm, Sweden
- City University of Hong Kong 2017–2021  
B.Eng. Mechanical Engineering Hong Kong SAR, China
- National University of Singapore 2020  
Academic Exchange Singapore

## Research Interests

- Cyber-Physical Systems
- Control & Dynamics
- Reinforcement Learning
- Heterogeneous Robots Collaboration

## Journal Publications

- K. Tan, X. Niu, Q. Ji, L. Feng, and M. Törngren. "Optimal gait design for a soft quadruped robot via multi-fidelity Bayesian optimization," *Applied Soft Computing*, vol. 169, p. 112568, 2025.

## Conference Publications

- J. Xu, X. Niu, D. G. Broo, K. Hjort. TouchDrive: Electronics-Free Tactile Sensing Interface for Assistive Grasping. Accepted for presentation at the RoboTac workshop, with IEEE International Conference on Robotics and Automation (ICRA) 2026.
- A. Rouchitsas, X. Niu, G. Castellano and D. G. Broo. "What do I do now?": Spontaneous Human Responses to Robot Effectiveness and Efficiency Malfunctions in Collaborative Robotics. In *The ACM Conference on Human Factors in Computing Systems (CHI)*. ACM, 2026.
- X. Niu and D. G. Broo. Investigating Symbiosis in Robotic Ecosystems: A Case Study for Multi-Robot Reinforcement Learning Reward Shaping. In *2025 9th International Conference on Robotics and Automation Sciences (ICRAS)*. IEEE, 2025.
- X. Niu, N. C. Barajas and D. G. Broo. Enabling Symbiosis in Multi-Robot Systems through Multi-Agent Reinforcement Learning. In *2025 IEEE 8th International Conference on Industrial Cyber-Physical Systems (ICPS)*. IEEE, 2025.
- X. Niu\*, K. Tan\*, D. G. Broo and L. Feng. Optimal Gait Control for a Tendon-driven Soft Quadruped Robot by Model-based Reinforcement Learning. In *2025 International Conference on Robotics and Automation (ICRA)*. IEEE, 2025.

## Working Papers

- X. Niu and D. G. Broo. PRISM: Policy-shaping via Reward decomposition for Inter-agent Symbiosis in Multi-Robot Cooperation. Under submission to journal publication.
- J. Xu, X. Niu, J. Andersson, D. G. Broo and K. Hjort. Miniaturized Multifunctional Valves for Intelligent Pneumatic Systems in Soft Robotics. Under submission to journal publication.
- X. Niu and D. G. Broo. MORPH: Self-Organising Multi-Robot Task Allocation via Neuroplasticity-Inspired Adaptive Topology. Under submission to conference.

## Other Publications

- Maser Thesis: X. Niu (2023). Optimal Gait Control of Soft Quadruped Robot by Model-based Reinforcement Learning. Thesis, 2023. Available: DiVA, id: diva2:1810127.
- HK project: C. Egenäs\*, F. Ekman\*, C. Ma\*, T. Naser\*, X. Niu\*, A. Sernelin\*, S. Stenow\*, and B. Ström\*, "Electronically Vacuum Regulated Shut-off Valve for Milking System," Report (Refereed), 2023. [Online]. Available: DiVA, id: diva2:1738909.

## Professional Service

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- Reviewer for Journal of Field Robotics (JFR), IEEE International Conference on Robotics and Automation (ICRA), IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), IEEE-RAS International Conference on Humanoid Robots (Humanoids), IEEE International Conference on Industrial Cyber-Physical Systems (ICPS), and IEEE International Conference on Robot and Human Interactive Communication (ROMAN).
- Teaching assistant for master level courses at KTH (MF2007) and Uppsala (1DT106, 1DT108, 1DT054, 1RT495, 1DT104, 1DT059)
- Master thesis supervision (Ibrahim Bala VT25, Liu Yang VT26) and Bachelor (Jonathan Näslund VT25)

## Awards & Achievements

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|-----------------------------------------------------------------------------------------------------------|---------|
| • IEEE Robotics and Automation Society Travel Grant Awardee for ICRA, Atlanta, United States              | 2025.5  |
| • Talent Development Scholarship, Hong Kong SAR, China                                                    | 2020.6  |
| • Second Prize in National Finals of the “Challenge Cup” Competition, Beijing, China                      | 2019.11 |
| • Silver Prize in National Finals of “Internet +” Competition, Hangzhou, China                            | 2019.10 |
| • Second Prize in HK University Student Innovation and Entrepreneurship Competition, Hong Kong SAR, China | 2019.4  |

## Skills

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MATLAB/Simulink, Python, C/C++, R, ROS/ROS2, MoveIt, PyTorch, OpenCV, Gazebo, Isaac Sim, Gym/Gymnasium, URDF/SDF/Xacro, RRT\*, PRM, A\*, Dijkstra, PID, MPC, adaptive,  $H^\infty$ , HJB, EKF, UKF, RL (PPO, SAC, DQN, DDPG), RGB-D/LiDAR perception, SLAM, Optical/Stereo cameras, IMU, Encoder, Strain Gauge, Force/Torque Sensor, Fluid/Air Pressure Sensor, motor (BLDC, PMSM, stepper, servo, H-bridge, FOC), STM32, ESP32, Jetson, Raspberry Pi, NXP LPC, Zephyr, FreeRTOS, Keil, UART, SPI, I<sup>2</sup>C, TCP/IP, Modbus, DDS, MQTT, SolidWorks, Solid Edge, AutoCAD, Autodesk EAGLE, KLayout, COMSOL, LS-DYNA, 3D prototyping, CNC machining, lithography, CVD, PVD, etching (RIE/DRIE), doping, SEM/TEM, Inkscape, L<sup>A</sup>T<sub>E</sub>X.